DV Assignment 2 Netflix Data-Analysis

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In [1]:
```
import pandas as pd
net=pd.read_csv("netflix.csv")
net
```

Out[1]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody	United States	November 1, 2019	2009

	show_id	type	title	director	cast	country	date_added	release_year
					Harrelson, Emma Stone, ...			
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

8807 rows × 12 columns

In [2]:
```
net.shape
```

Out[2]:
```
(8807, 12)
```

In [3]:
```
net.describe()
```

Out[3]:

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

In [4]:
```
net.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

In [5]:
```
net.isnull().sum()
```

Out[5]:
```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```

In [6]:
```
net["director"].fillna("Unknown",inplace=True)
```

In [7]:
```
net["cast"].fillna("Not Available",inplace=True)
```

In [8]:
```
net["country"].fillna("Unknown",inplace=True)
```

In [9]:
```
net["date_added"].fillna("NaT",inplace=True)
```

In [10]:
```
net["rating"].fillna("Not Rated",inplace=True)
```

In [11]:
```
net["director"].isnull().sum()
```

Out[11]:
```
np.int64(0)
```

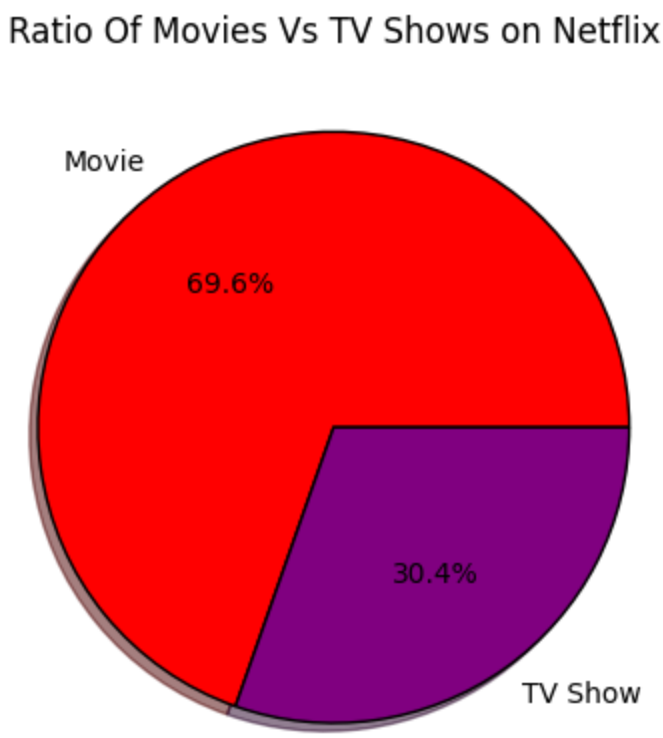
In [12]:
```
net["duration"].fillna(1,inplace=True)
```

In [13]:
```
net.isnull().sum()
```

Out[13]:
```
show_id         0
type            0
title           0
director        0
cast            0
country         0
date_added      0
release_year    0
rating          0
duration        0
listed_in       0
description     0
dtype: int64
```

1. What is the ratio of Movies vs TV Shows on Netflix?

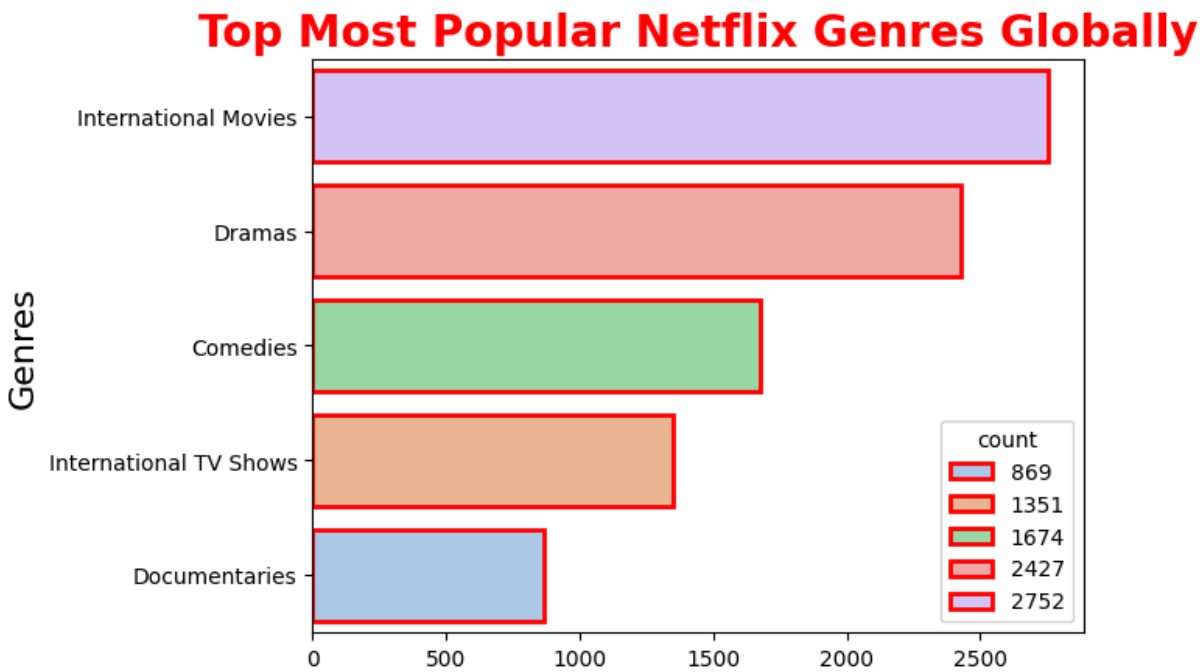
In [46]:
```
ratio_movie=net["type"].value_counts()
import matplotlib.pyplot as plt
plt.pie(x=ratio_movie,labels=ratio_movie.index,autopct="%1.1f%%",colors=["red","pur
plt.title("Ratio Of Movies Vs TV Shows on Netflix")
plt.show()
```



Insight: Movies vs. TV Shows Ratio: The platform is significantly tilted towards cinematic content. Out of 8,807 total titles, there are 69% Movies compared to 30% TV Shows. This indicates a historical priority on high-volume movie acquisitions, though the engagement metrics often favor the recurring nature of shows.

2. Which genres are most popular on Netflix globally?

In [49]:
```
import matplotlib.pyplot as plt
import seaborn as sb
genres=net["listed_in"].str.split(",").explode().str.strip()
top_genres=genres.value_counts().sort_values(ascending=False).head()
top_genres
sb.barplot(x=top_genres.values,y=top_genres.index,hue=top_genres,palette="pastel",e
plt.title("Top Most Popular Netflix Genres Globally",fontsize=20,color="red",fontwe
plt.ylabel("Genres",fontsize=16)
plt.show()
```



insights: Across the globe, international movies, dramas, and comedies are the most popular genres. This shows that audiences really connect with stories that feel local and emotionally relatable, highlighting a clear direction for future content acquisition

In [16]:
```
net["release_year"].info()
```

```
<class 'pandas.core.series.Series'>
RangeIndex: 8807 entries, 0 to 8806
Series name: release_year
Non-Null Count  Dtype
--------------  -----
8807 non-null   int64
dtypes: int64(1)
memory usage: 68.9 KB
```

In [17]:
```
net.groupby("release_year").count()
```

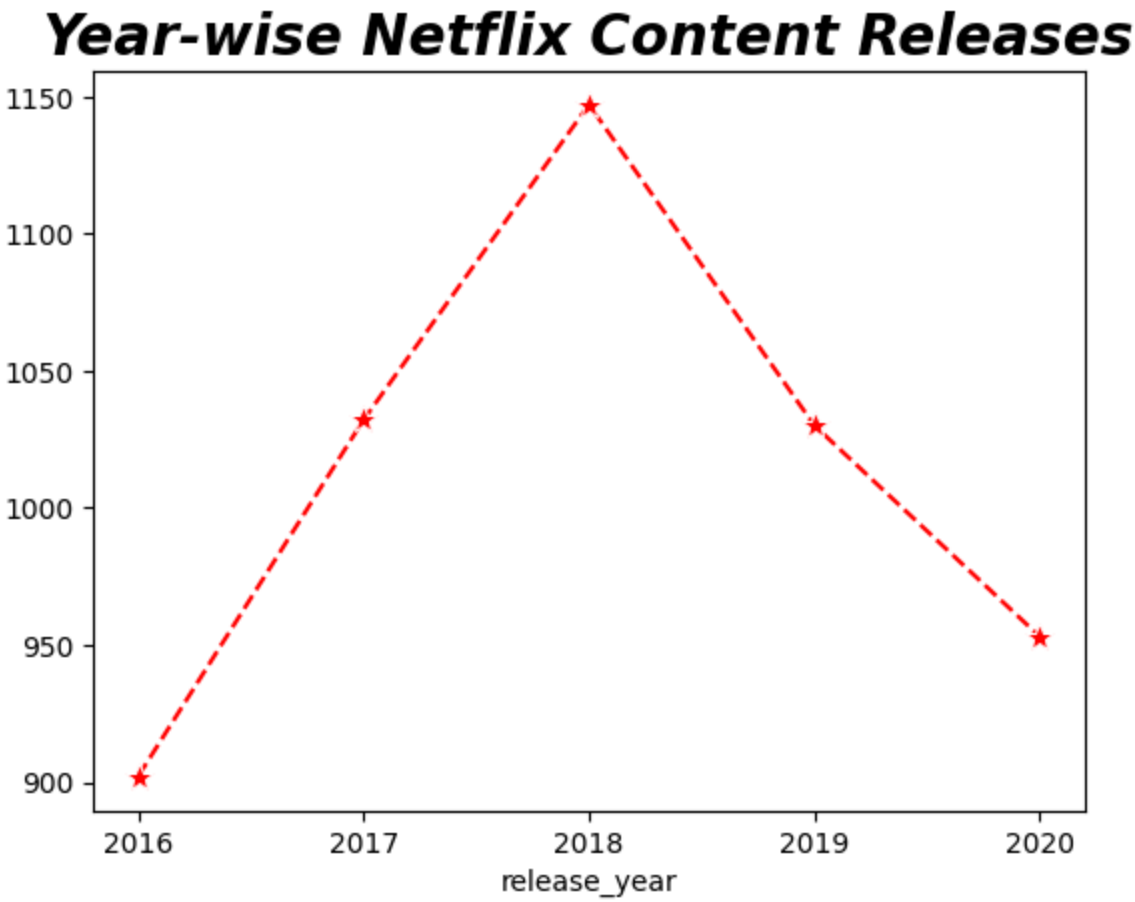
Out[17]:

release_year	show_id	type	title	director	cast	country	date_added	rating	duration
1925	1	1	1	1	1	1	1	1	1
1942	2	2	2	2	2	2	2	2	2
1943	3	3	3	3	3	3	3	3	3
1944	3	3	3	3	3	3	3	3	3
1945	4	4	4	4	4	4	4	4	4
...	...	...	...	...	...	...	...	...	...
2017	1032	1032	1032	1032	1032	1032	1032	1032	1032
2018	1147	1147	1147	1147	1147	1147	1147	1147	1147
2019	1030	1030	1030	1030	1030	1030	1030	1030	1030
2020	953	953	953	953	953	953	953	953	953
2021	592	592	592	592	592	592	592	592	592

74 rows × 11 columns

3. Which years saw the highest release of content on Netflix?

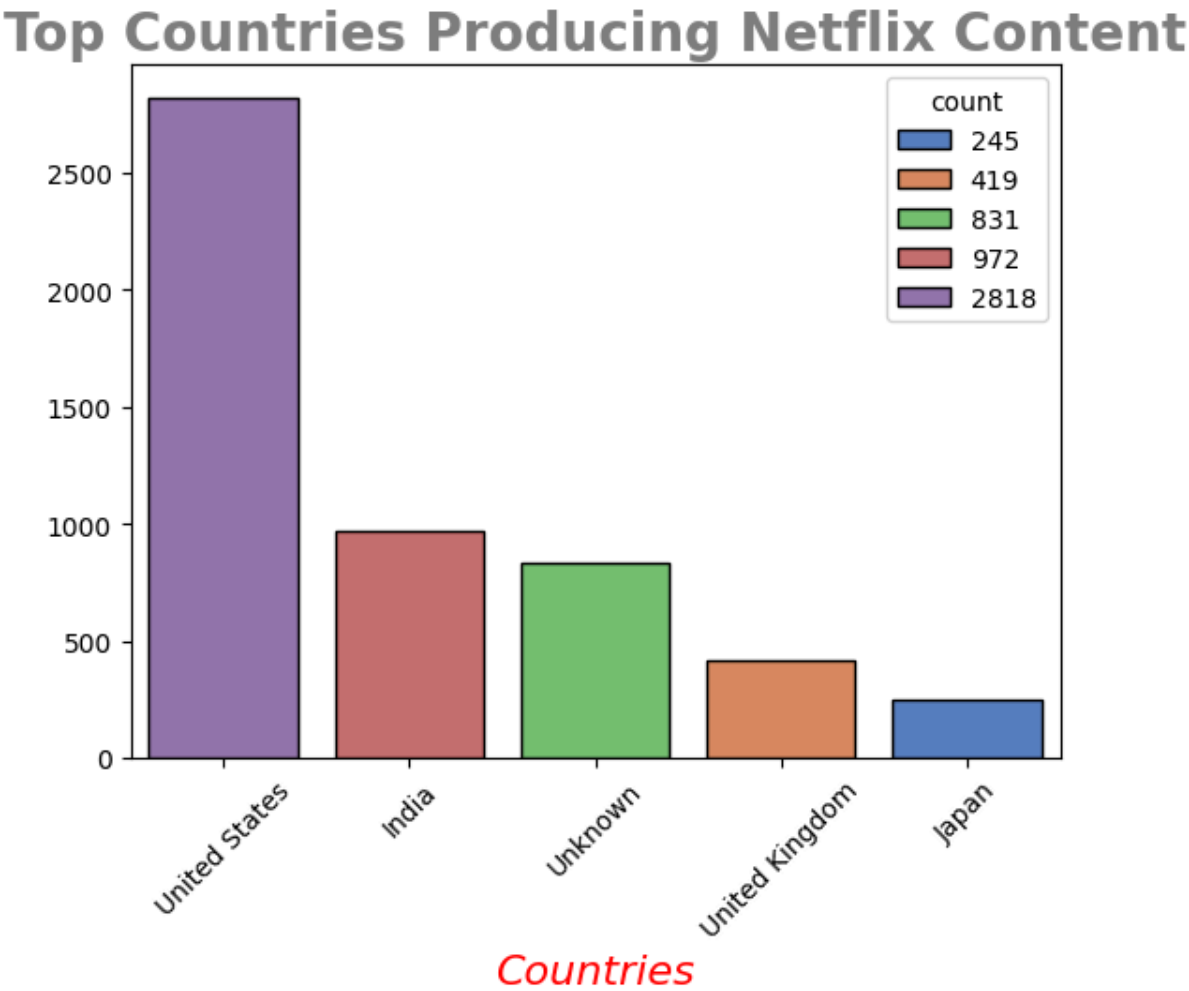
In [18]:
```
import matplotlib.pyplot as plt
import seaborn as sb
hrc=net["release_year"].value_counts().sort_values(ascending=False).head()
sb.lineplot(x=hrc.index,y=hrc.values,marker="*",color="red",markersize=10,linestyle
plt.xticks(range(2016, 2021))
plt.title("Year-wise Netflix Content Releases",fontsize=20,fontstyle="italic",fontw
plt.show()
```



Insight : Netflix experienced a big surge in new releases between 2017 and 2019, peaking in 2018 and 2019. Release activity slowed down a bit in 2020 and 2021.

4. Which countries produce the most Netflix content?

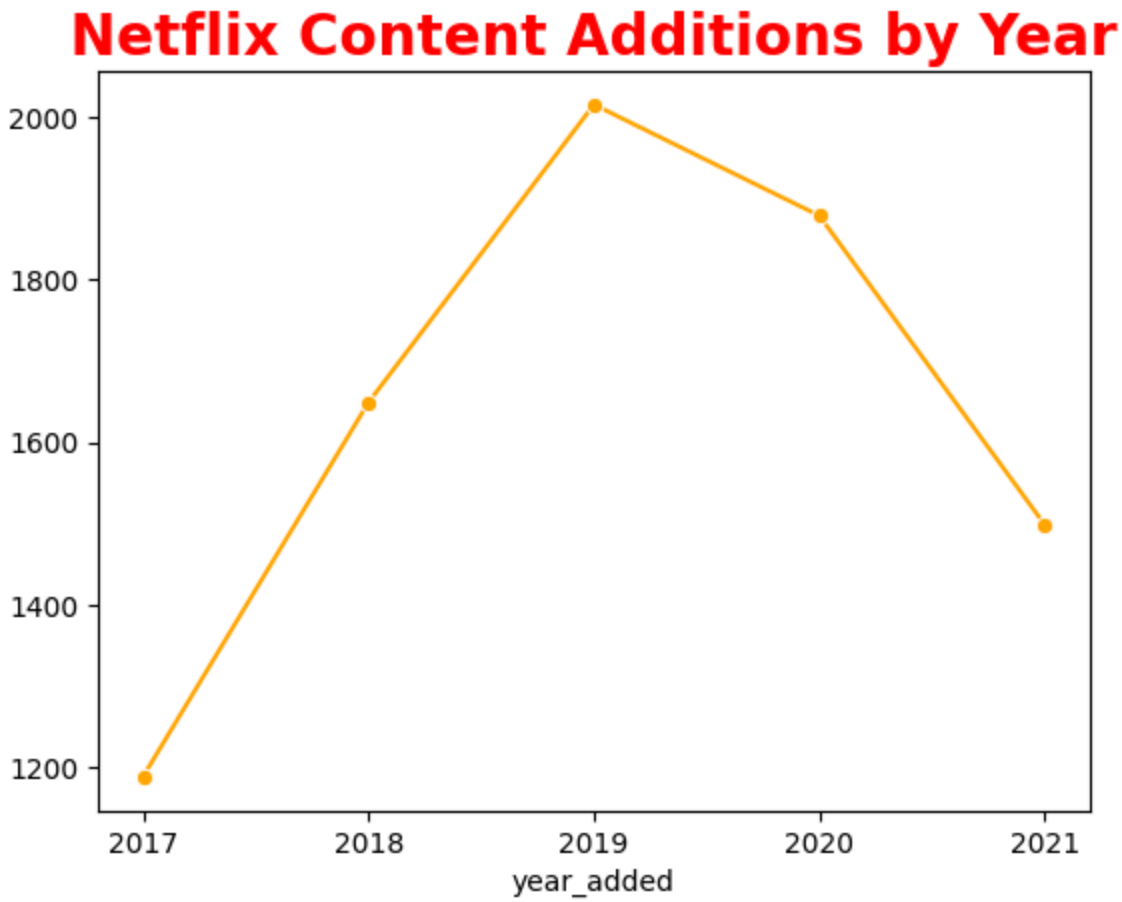
In [19]:
```
import matplotlib.pyplot as plt
import seaborn as sb
most_content_by_countries=net["country"].value_counts().sort_values(ascending=False
sb.barplot(x=most_content_by_countries.index,y=most_content_by_countries.values,hue
plt.xticks(rotation=45)
plt.title("Top Countries Producing Netflix Content",color="gray",fontsize=20,fontwe
plt.xlabel("Countries",color="red",fontsize=16,fontstyle="italic")
plt.show()
```



Insight : The United States leads as the biggest content producer on Netflix, followed by India and the United Kingdom. These three markets continue to drive much of the platform's global appeal

5. How has the trend of adding new content evolved year by year?

In [20]:
```
import matplotlib.pyplot as plt
import seaborn as sb
net["date_added"] = net['date_added'].astype(str).str.strip()
net["date_added"] = pd.to_datetime(net['date_added'], errors='coerce')
net["year_added"] = net['date_added'].dt.year
trend=net["year_added"].value_counts().sort_values(ascending=False).head()
sb.lineplot(x=trend.index,y=trend.values,marker="o",linestyle="-",color="orange")
plt.xticks(range(2017,2022))
plt.title("Netflix Content Additions by Year",fontsize=20,color="red",fontweight="b
plt.show()
```

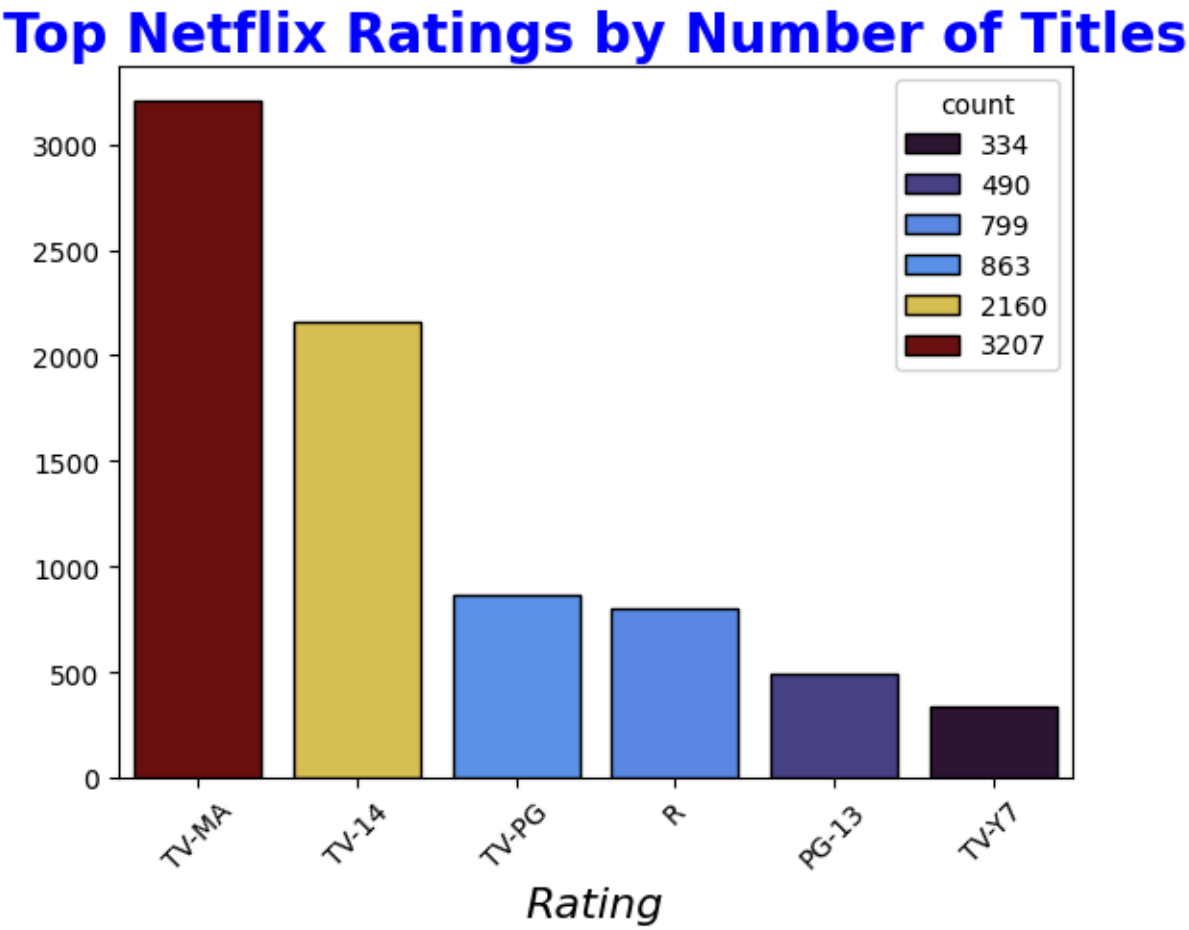


Insight : Netflix's content library began growing rapidly around 2015, hitting its peak in 2019. This growth reflects the company's shift from relying mostly on licensed titles to becoming a major creator of original 'Netflix Originals.

6. Which ratings (e.g., TV-MA, PG, etc.) are most frequent on Netflix?

In [21]:
```
import matplotlib.pyplot as plt
import seaborn as sb
rating=net["rating"].value_counts().sort_values(ascending=False).head(6)
sb.barplot(x=rating.index,y=rating.values,hue=rating,palette="turbo",edgecolor="bla
plt.xticks(rotation=45)
plt.xlabel("Rating",fontsize=16,fontstyle="italic")
plt.title("Top Netflix Ratings by Number of Titles",fontsize=20,color="blue",fontwe
```

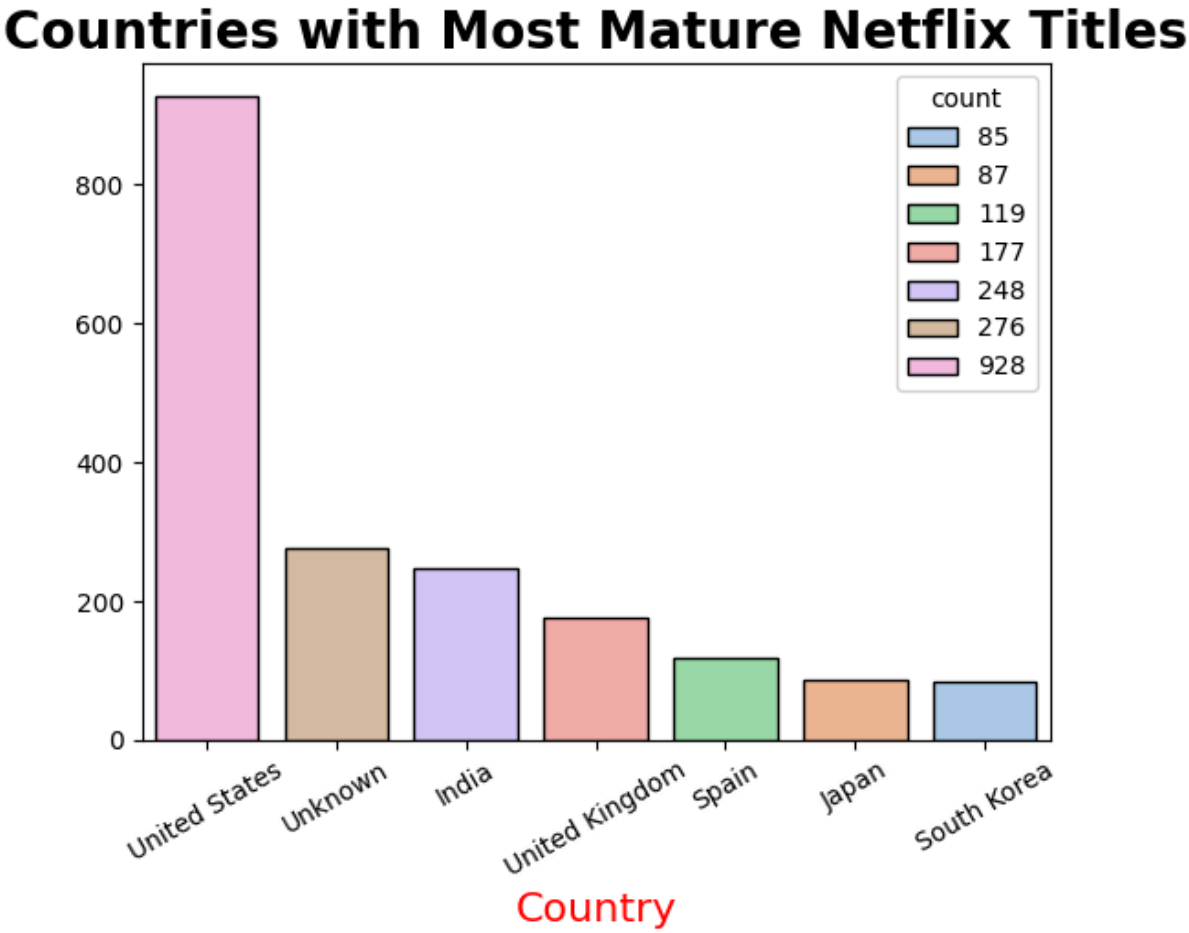
Out[21]:
```
Text(0.5, 1.0, 'Top Netflix Ratings by Number of Titles')
```



Insight : Most Netflix content is rated TV-MA, with TV-14 coming in next. This shows the platform focuses largely on adult and teen audiences, rather than programming for young children

7. Do some countries tend to produce more mature content (TV-MA)?

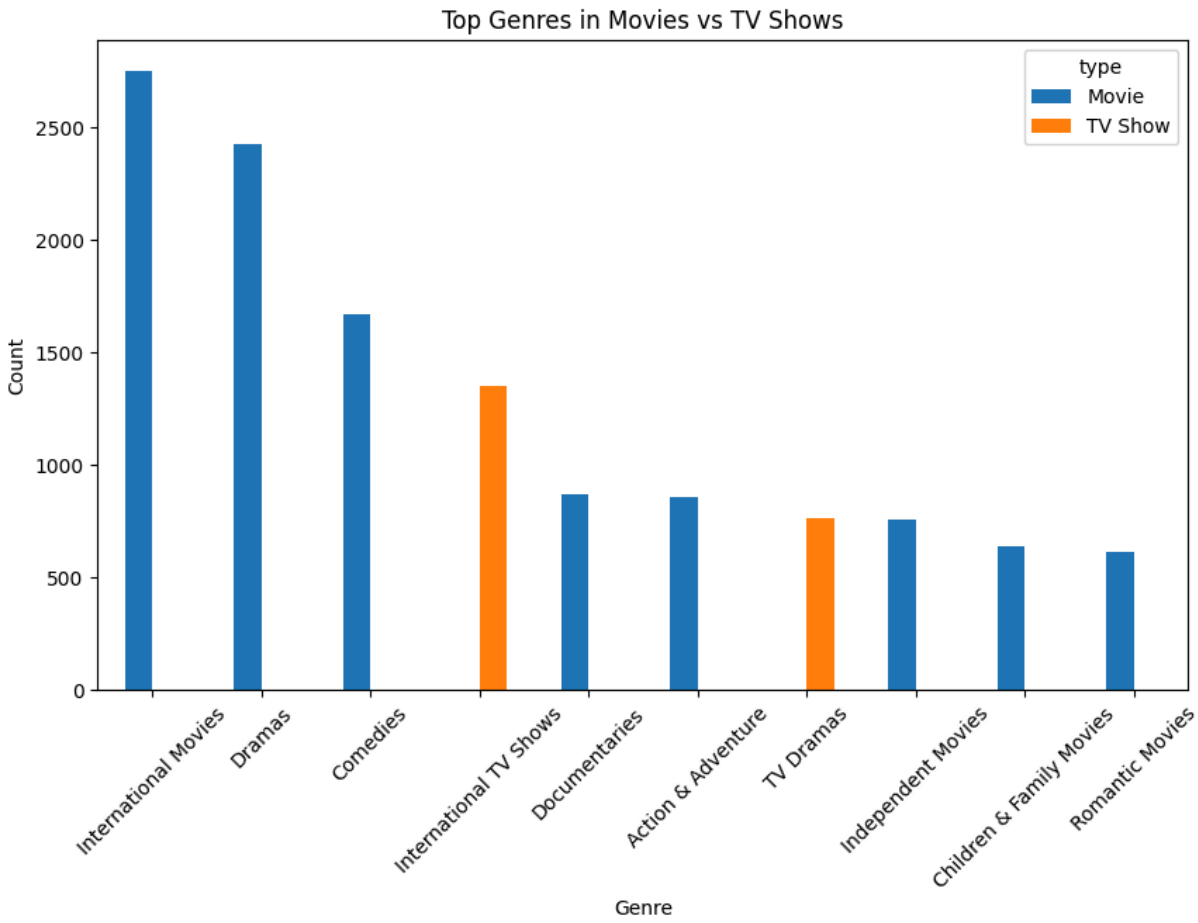
In [22]:
```
mature_content=net[net["rating"]=="TV-MA"]
mature_country=mature_content["country"].value_counts(ascending=False).head(7)
import matplotlib.pyplot as plt
import seaborn as sb
sb.barplot(x=mature_country.index,y=mature_country.values,hue=mature_country,edgeco
plt.xticks(rotation=30)
plt.title("Countries with Most Mature Netflix Titles",fontsize=20,fontweight="bold"
plt.xlabel("Country",color="red",fontsize=16)
plt.show()
```



Insight : Countries such as the United States, United Kingdom, and South Korea produce a high volume of TV-MA content. This is important for tailoring regional marketing strategies around edgy dramas and sophisticated thrillers.

8. Which genres are more associated with TV Shows vs Movies?

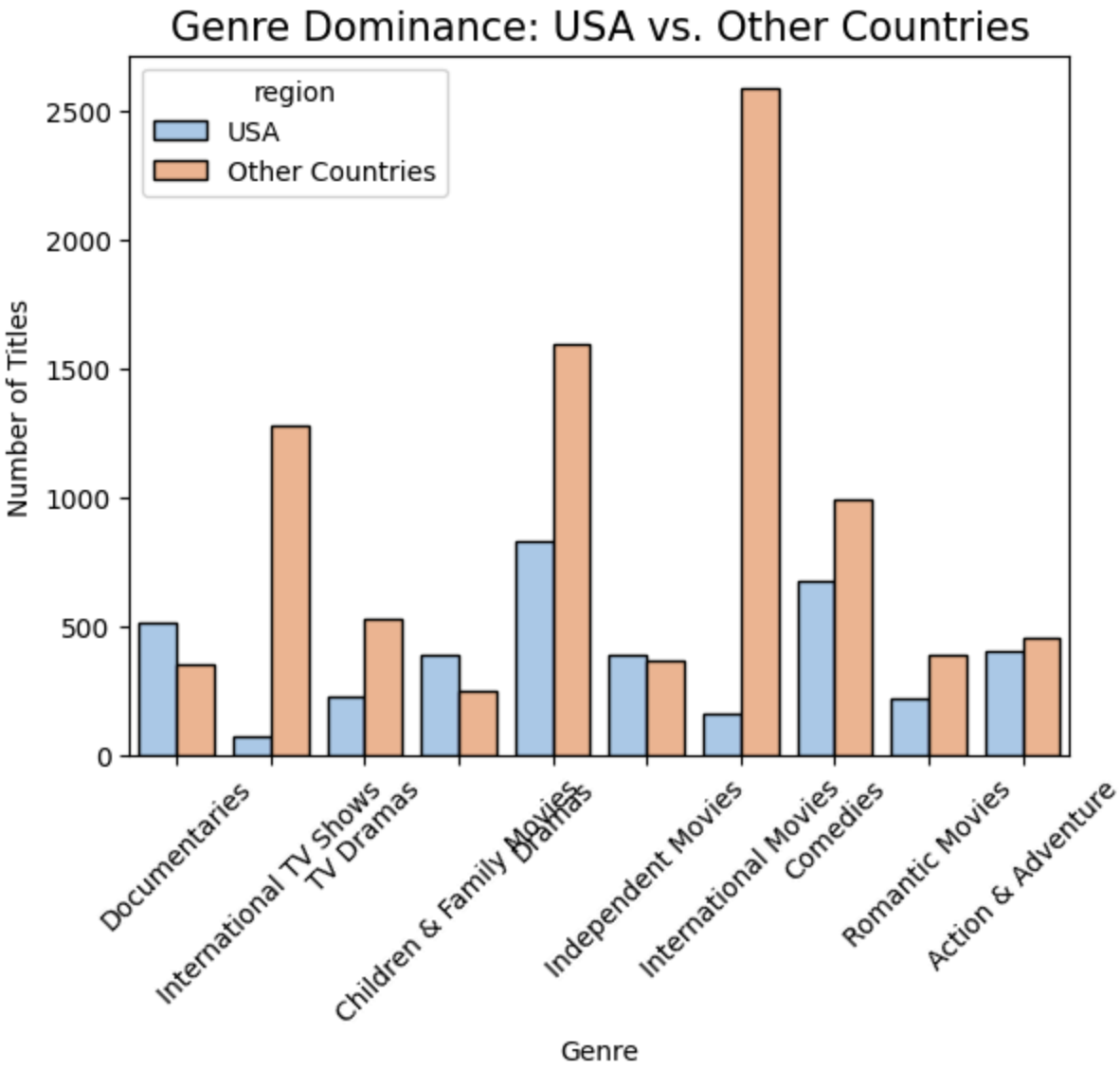
In [23]:
```
net_genre = net.assign(genre=net['listed_in'].str.split(', ')).explode('genre')
genre_type = net_genre.groupby(['type','genre']).size().reset_index(name='count')
genre_pivot = genre_type.pivot(index='genre', columns='type', values='count').filln
genre_pivot.sort_values(by='Movie', ascending=False).head(10)
import matplotlib.pyplot as plt
top_genres = genre_pivot.sum(axis=1).sort_values(ascending=False).head(10).index
genre_pivot.loc[top_genres].plot(kind='bar', figsize=(10,6))
plt.title("Top Genres in Movies vs TV Shows")
plt.xlabel("Genre")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```



Insight : Dramas and international movies lead the movie category, while international TV shows and TV dramas form the core of the TV show lineup.

9. Which genres dominate the U.S. vs other countries?

In [24]:
```
import matplotlib.pyplot as plt
import seaborn as sb
net['region'] = 'Other Countries'
net.loc[net['country'].str.contains('United States', na=False), 'region'] = 'USA'
net['genre_list'] = net['listed_in'].str.split(', ')
genre_df = net.explode('genre_list')
top_10_genres = genre_df['genre_list'].value_counts().head(10).index
final_df = genre_df[genre_df['genre_list'].isin(top_10_genres)]
sb.countplot(data=final_df, x="genre_list", hue="region", palette="pastel",edgecolo
plt.title("Genre Dominance: USA vs. Other Countries", fontsize=15)
plt.xlabel("Genre")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```



Insight : In the U.S., Netflix features a higher concentration of documentaries and stand-up comedy, while international markets are largely driven by romance and action titles.

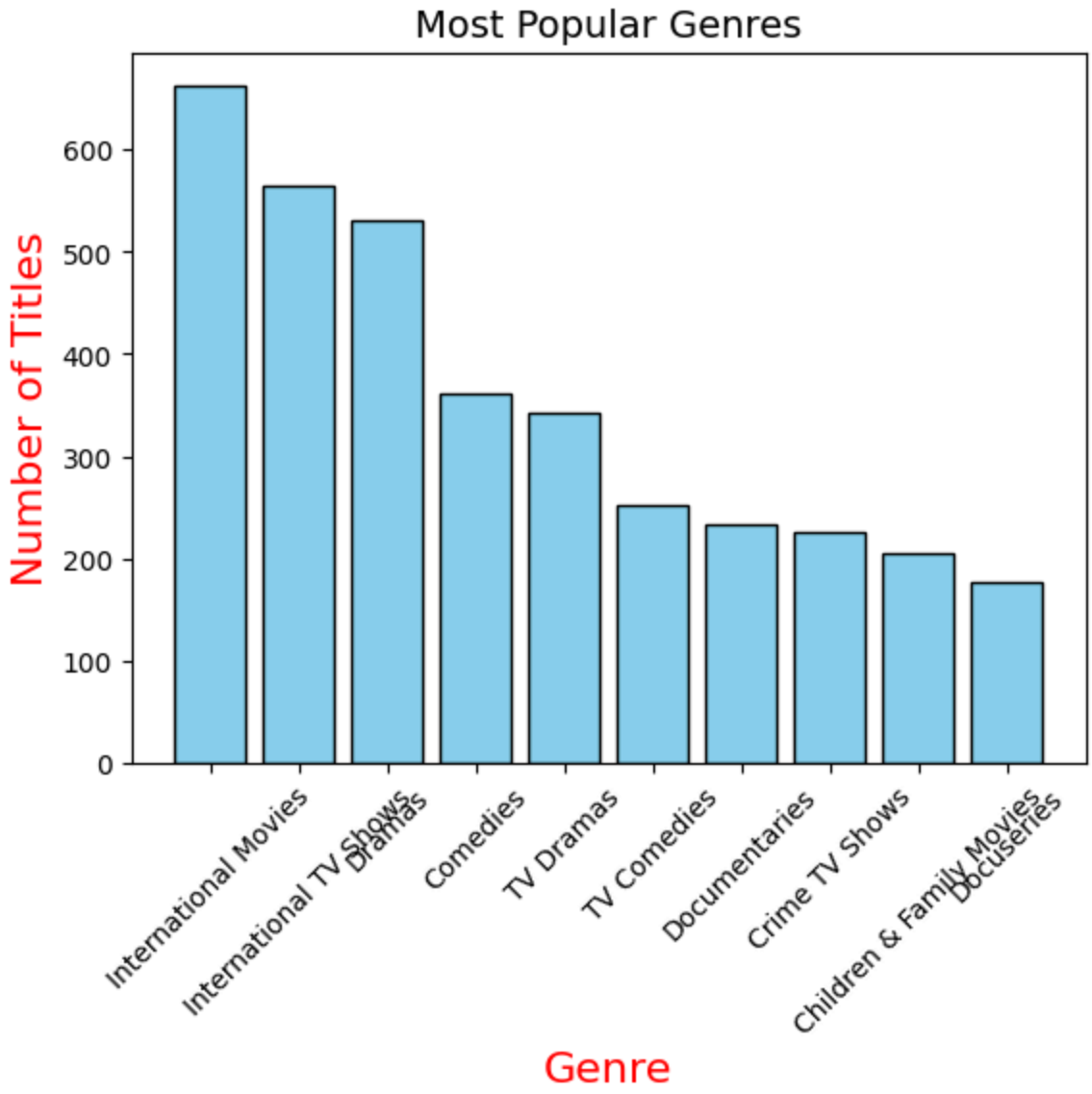
10. What genres are most popular in the last 3 years?

In [25]:
```
import matplotlib.pyplot as plt
year = net['release_year'].max()
year =year - 2
recent_net = net[net['release_year'] >=year]
recent_net['genre_list'] = recent_net['listed_in'].str.split(', ')
genre_df = recent_net.explode('genre_list')
top_10 = genre_df['genre_list'].value_counts().head(10)
plt.bar(top_10.index, top_10.values, color='skyblue', edgecolor='black')
plt.title("Most Popular Genres",fontsize=14)
plt.ylabel('Number of Titles',fontsize=16,color="red")
plt.xlabel("Genre",fontsize=16,color="red")
```

```
plt.xticks(rotation=45)
plt.show()
```

```
C:\Users\ASUS\AppData\Local\Temp\ipykernel_10648\2488932064.py:5: SettingWithCopyWar
ning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u
ser_guide/indexing.html#returning-a-view-versus-a-copy
  recent_net['genre_list'] = recent_net['listed_in'].str.split(', ')
```



Insight : Between 2019 and 2021, documentaries and international TV shows saw significant growth, reflecting a global audience that’s increasingly open to subtitles and drawn to true-crime and real-life stories.

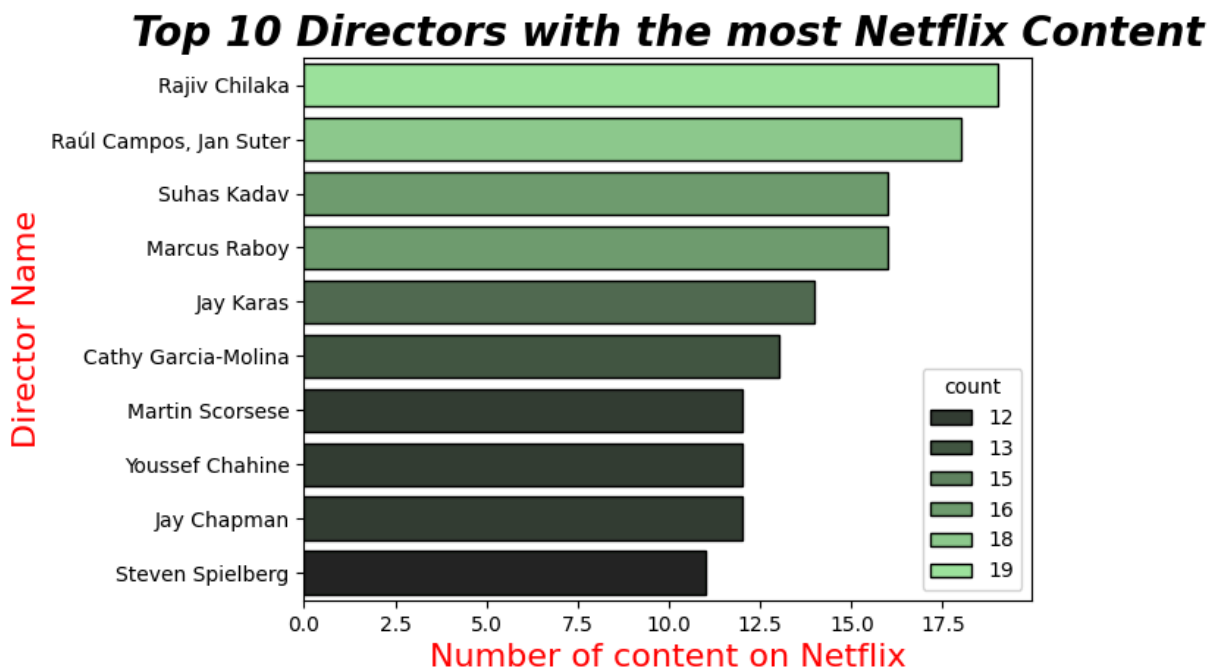
11. Who are the top 10 directors with the most Netflix content?

In [26]:
```
top_directors=net[net["director"]!="Unknown"]["director"].value_counts().sort_value
import matplotlib.pyplot as plt
import seaborn as sb
sb.barplot(x=top_directors.values,y=top_directors.index,hue=top_directors,color="li
plt.xlabel("Number of content on Netflix",fontsize=16,color="red")
plt.ylabel("Director Name",fontsize=16,color="red")
plt.title("Top 10 Directors with the most Netflix Content",fontsize=20,fontstyle="i
plt.show()
```

```
C:\Users\ASUS\AppData\Local\Temp\ipykernel_10648\2519478841.py:4: FutureWarning:

Setting a gradient palette using color= is deprecated and will be removed in v0.14.
0. Set `palette='dark:lightgreen'` for the same effect.

  sb.barplot(x=top_directors.values,y=top_directors.index,hue=top_directors,color="l
ightgreen",edgecolor="black")
```



Insight : Directors such as Rajiv Chilaka, known for kids' animation, and Jan Suter are among the most prolific on Netflix. Collaborating with high-output creators helps secure a steady stream of 'anchor' content for the platform

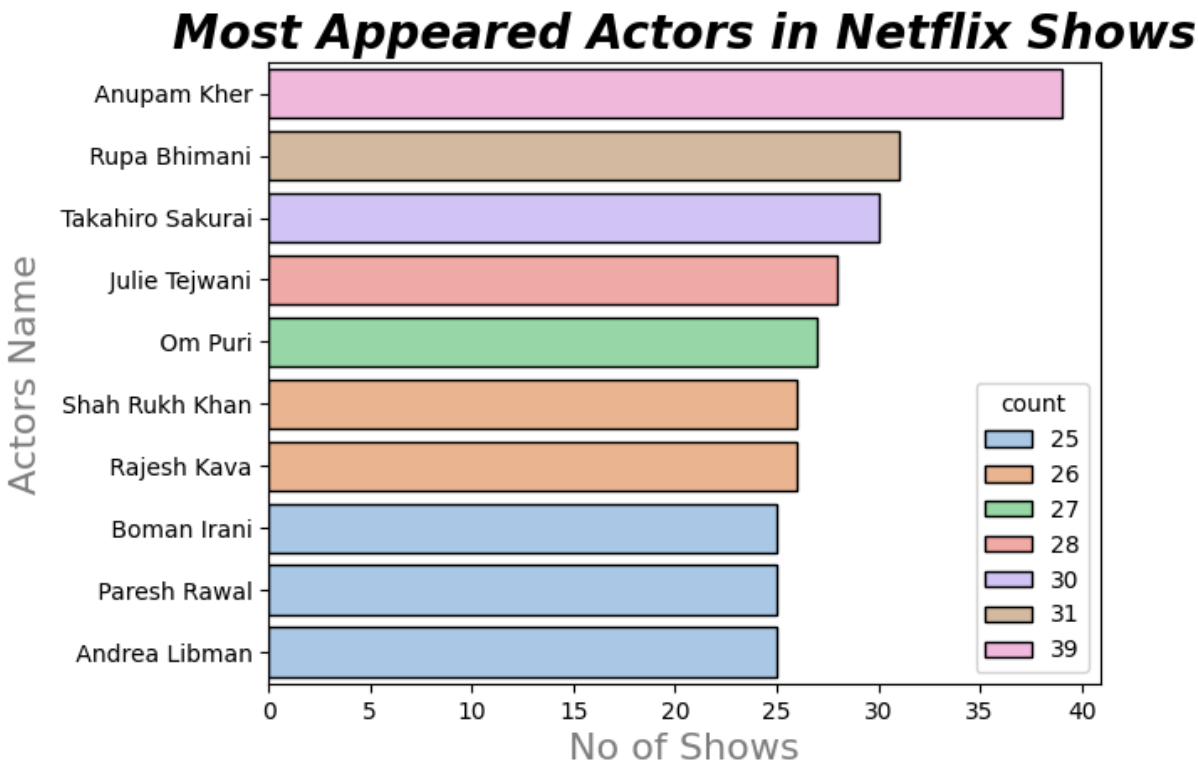
In [27]:
```
net["cast"].value_counts()
```

Out[27]:
```
cast
Not Available
825
David Attenborough
19
Vatsal Dubey, Julie Tejwani, Rupa Bhimani, Jigna Bhardwaj, Rajesh Kava, Mousam, Sw
apnil                                                                          14
Samuel West
10
Jeff Dunham
7

...
Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, Mansoor Alfeeli, Ahd
1
Mark Ruffalo, Jake Gyllenhaal, Robert Downey Jr., Anthony Edwards, Brian Cox, Elia
s Koteas, Donal Logue, John Carroll Lynch, Dermot Mulroney, Chloë Sevigny      1
Jesse Eisenberg, Woody Harrelson, Emma Stone, Abigail Breslin, Amber Heard, Bill M
urray, Derek Graf                                                              1
Tim Allen, Courteney Cox, Chevy Chase, Kate Mara, Ryan Newman, Michael Cassidy, Sp
encer Breslin, Rip Torn, Kevin Zegers                                          1
Ankush Choudhary, Smita Shewale, Mohan Joshi, Atmaram Bhende, Asha Bhende
1
Name: count, Length: 7693, dtype: int64
```

12. Which actors appear most frequently in Netflix shows?

In [28]:
```
top_actor=net[net["cast"]!="Not Available"]["cast"].str.split(",").explode().value_
# top_actor
import matplotlib.pyplot as plt
import seaborn as sb
sb.barplot(x=top_actor.values,y=top_actor.index,hue=top_actor,palette="pastel",edge
plt.xlabel("No of Shows",fontsize=16,color="gray")
plt.ylabel("Actors Name",fontsize=16,color="gray")
plt.title("Most Appeared Actors in Netflix Shows",fontsize=20,fontweight="bold",fon
plt.show()
```



Insight : Actors from the Indian film industry, such as Anupam Kher and Shah Rukh Khan, appear in a large number of Netflix titles. This underscores the platform's extensive Bollywood library

In [29]:
```
net.groupby("director")["listed_in"].count()
```

Out[29]:
```
director
A. L. Vijay            2
A. Raajdheep           1
A. Salaam              1
A.R. Murugadoss        2
Aadish Keluskar        1
                      ..
Çagan Irmak            1
Ísold Uggadóttir       1
Óskar Thór Axelsson    1
Ömer Faruk Sorak       2
Şenol Sönmez           2
Name: listed_in, Length: 4529, dtype: int64
```

In [30]:
```
net["director"].value_counts().sort_values(ascending=False).head()
```

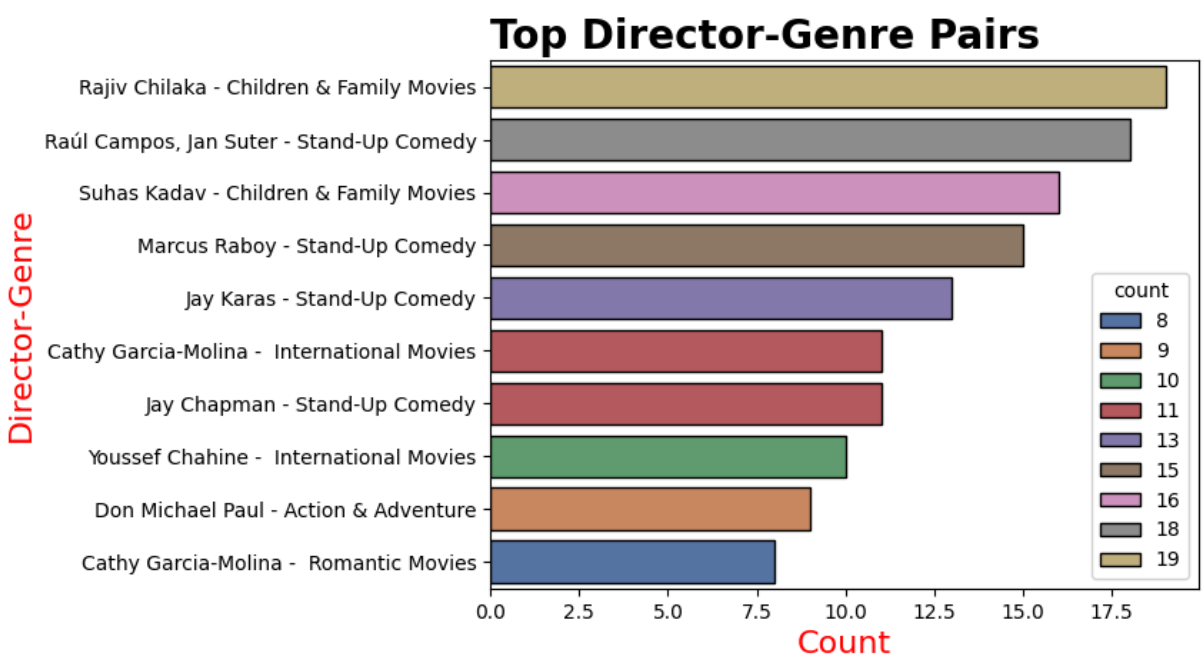
Out[30]:
```
director
Unknown                   2634
Rajiv Chilaka               19
Raúl Campos, Jan Suter      18
Suhas Kadav                 16
Marcus Raboy                16
Name: count, dtype: int64
```

In [31]:
```
top_directors = net["director"].dropna().value_counts().head(10)
print(top_directors)
```

```
director
Unknown                   2634
Rajiv Chilaka               19
Raúl Campos, Jan Suter      18
Suhas Kadav                 16
Marcus Raboy                16
Jay Karas                   14
Cathy Garcia-Molina         13
Martin Scorsese             12
Youssef Chahine             12
Jay Chapman                 12
Name: count, dtype: int64
```

13. Which director-genre pairs are most frequent?

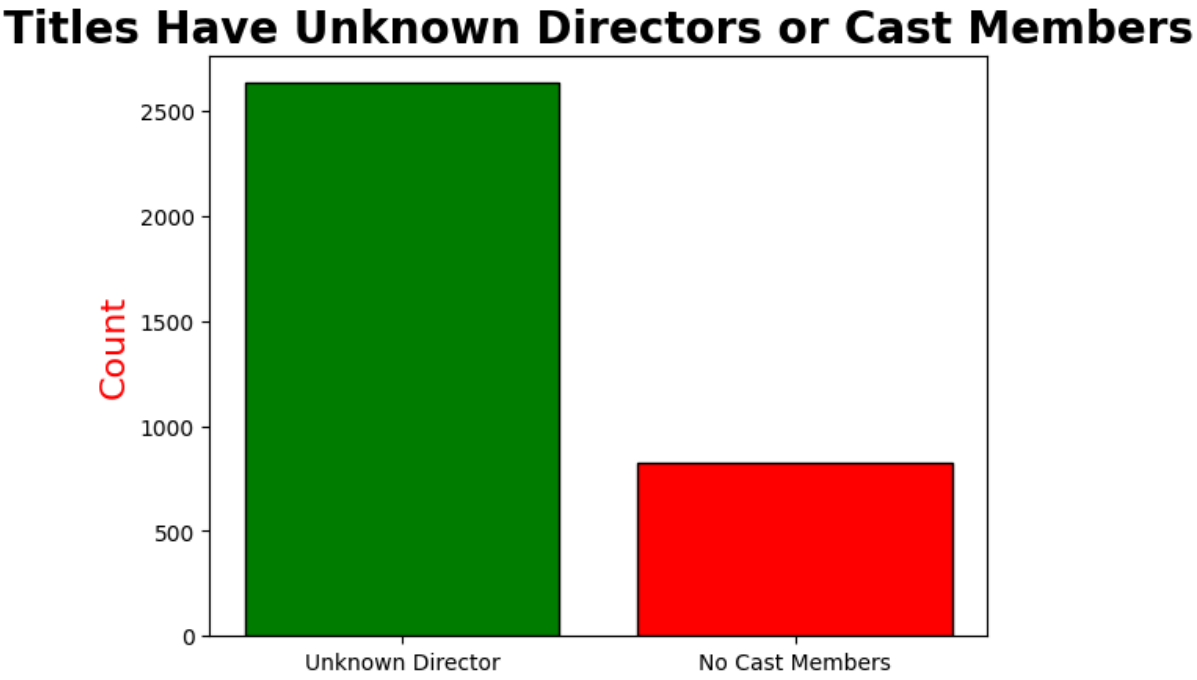
In [32]:
```
genres=net.assign(genre=net["listed_in"].str.split(",")).explode("genre")
genres=genres[genres["director"]!="Unknown"]
most_frequent=genres.groupby(["director","genre"]).size().reset_index(name="count")
pairs=most_frequent.sort_values(by="count",ascending=False).head(10)
import matplotlib.pyplot as plt
import seaborn as sb
pairs["label"] = pairs["director"] + " - " + pairs["genre"]
sb.barplot(y=pairs["label"],x= pairs["count"],hue=pairs["count"],palette="deep",edg
plt.xlabel("Count",fontsize=16,color="red")
plt.ylabel("Director-Genre",fontsize=16,color="red")
plt.title("Top Director-Genre Pairs",fontsize=20,fontweight="bold",loc="left")
plt.show()
```



Insight : Netflix shows a clear pattern of pairing certain directors with specific genres, like comedy and children's/family movies. This indicates the platform benefits from keeping proven creators within the genres where they excel

14. How many titles have unknown directors or cast members?

In [33]:
```
un_director=(net["director"]=="Unknown").sum()
no_cast=(net["cast"]=="Not Available").sum()
import matplotlib.pyplot as plt
import seaborn as sb
values=[un_director,no_cast]
labels=["Unknown Director","No Cast Members"]
plt.bar(labels,values,color=["green","red"],edgecolor="black")
plt.ylabel("Count",fontsize=16,color="red")
plt.title("Titles Have Unknown Directors or Cast Members",fontsize=20,fontweight="b
plt.show()
```



Insight : Data Gaps: A significant portion of the data (roughly 30%) had missing directors. Filling these gaps is essential for better recommendation algorithms and talent tracking.

In [34]:
```
net.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 15 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      8807 non-null   object
 4   cast          8807 non-null   object
 5   country       8807 non-null   object
 6   date_added    8797 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64
 8   rating        8807 non-null   object
 9   duration      8807 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
 12  year_added    8797 non-null   float64
 13  region        8807 non-null   object
 14  genre_list    8807 non-null   object
dtypes: datetime64[ns](1), float64(1), int64(1), object(12)
memory usage: 1.0+ MB
```

In [35]:
```
net["duration"].isnull().sum()
```

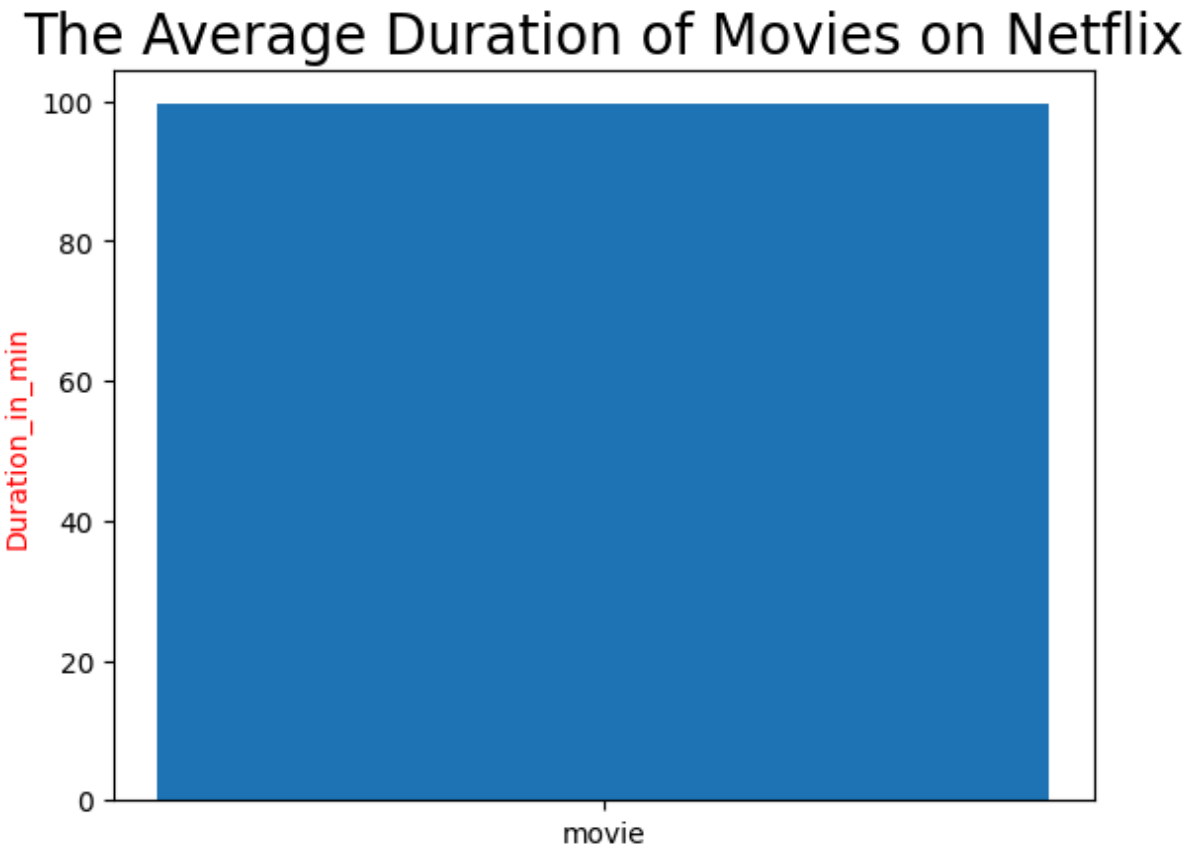
Out[35]:
```
np.int64(0)
```

In [36]:
```
net["type"].value_counts()
```

Out[36]:
```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```

15. What is the average duration of Movies on Netflix?

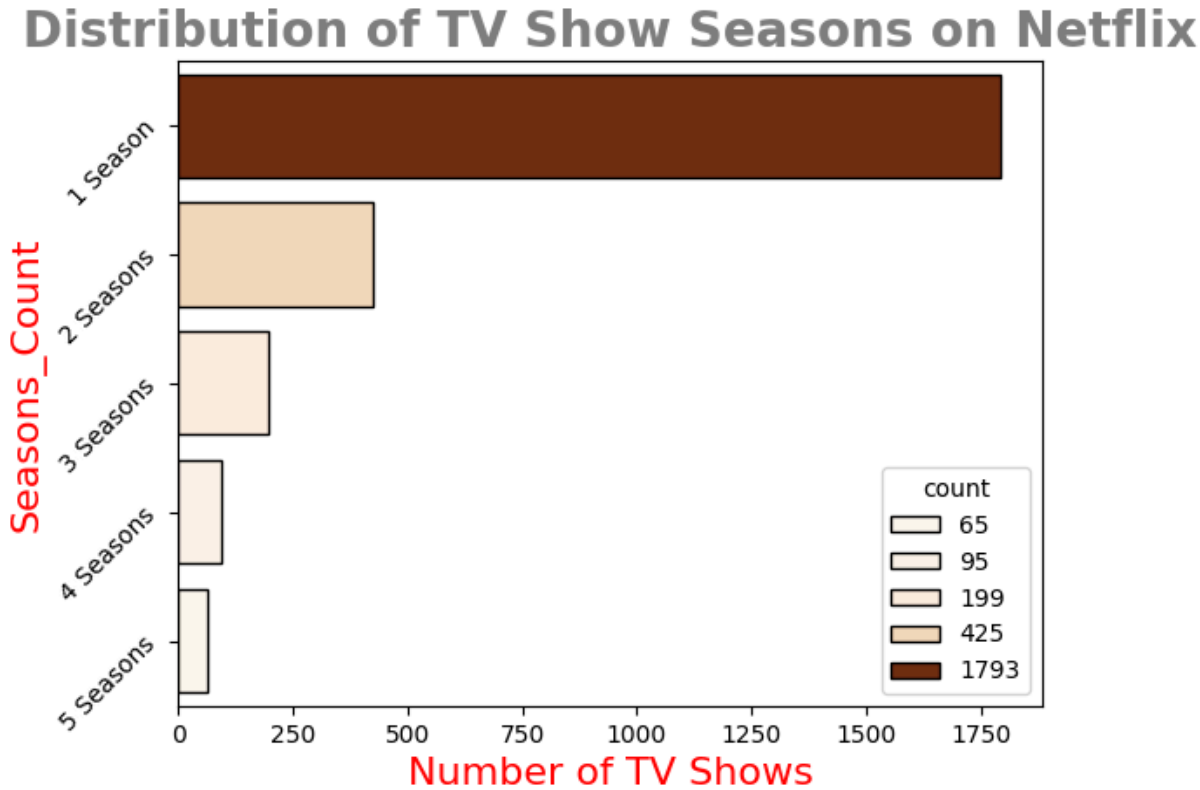
In [37]:
```
movie=net[net["type"]=="Movie"].copy()
movie["duration"]=movie["duration"].str.replace("min"," ")
movie["duration"]=movie["duration"].astype(float)
avg_duraction=movie["duration"].mean()
import matplotlib.pyplot as plt
import seaborn as sb
plt.bar(["movie"],[avg_duraction])
plt.ylabel("Duration_in_min",color="red")
plt.title("The Average Duration of Movies on Netflix",fontsize=20)
plt.show()
```



Insight: Netflix movies typically run between 90 and 110 minutes. This 'sweet spot' seems ideal for keeping viewers engaged without feeling too long.

16. What's the most common number of seasons for TV shows?

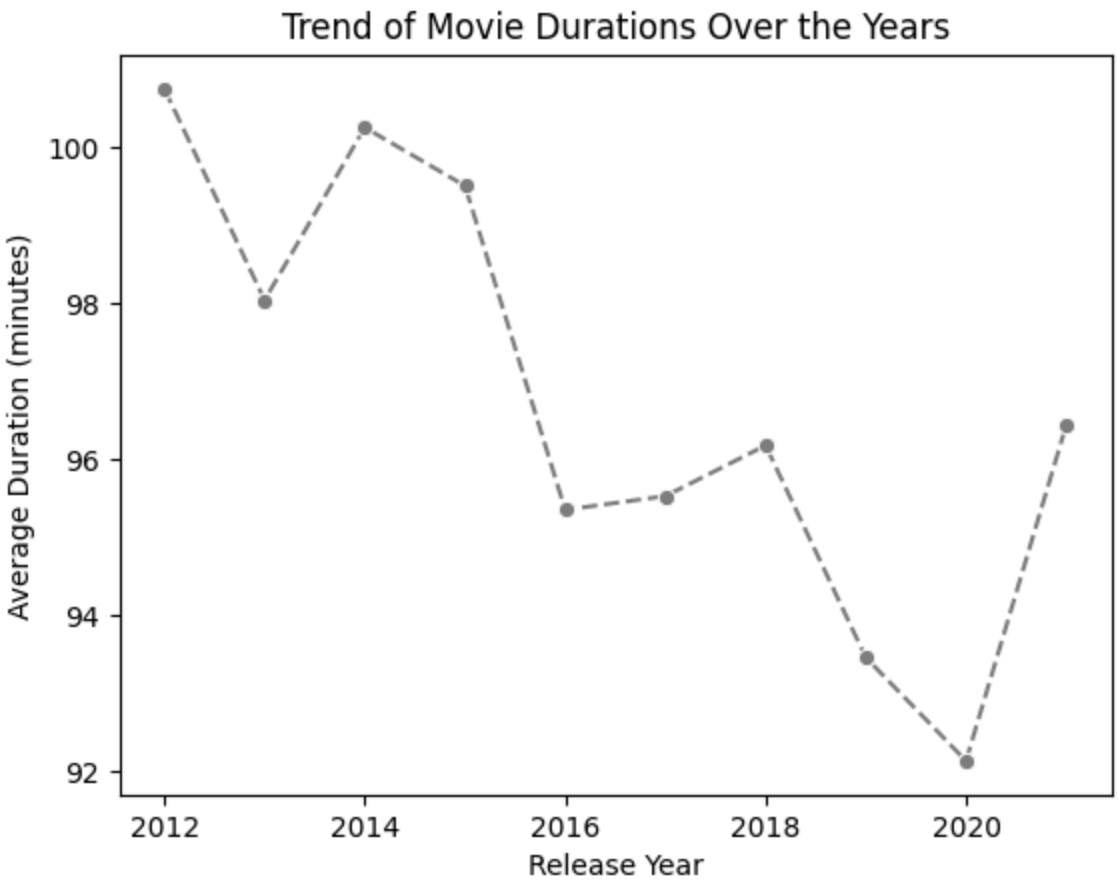
In [38]:
```
tv_show=net[net["type"]=="TV Show"]
common_season=tv_show["duration"].value_counts().sort_values(ascending=False).head(
import matplotlib.pyplot as plt
import seaborn as sb
sb.barplot(x=common_season.values,y=common_season.index,hue=common_season,palette="
plt.xlabel("Number of TV Shows",color="red",fontsize=16)
plt.ylabel("Seasons_Count",color="red",fontsize=16)
plt.title("Distribution of TV Show Seasons on Netflix",fontsize=20,color="gray",fon
plt.yticks(rotation=45)
plt.show()
```



Insight : Most Netflix TV shows have just one season, highlighting a 'one-and-done' trend. Many are either limited series or get canceled if they don't achieve quick success

17. Is there a trend in movie durations over the years?

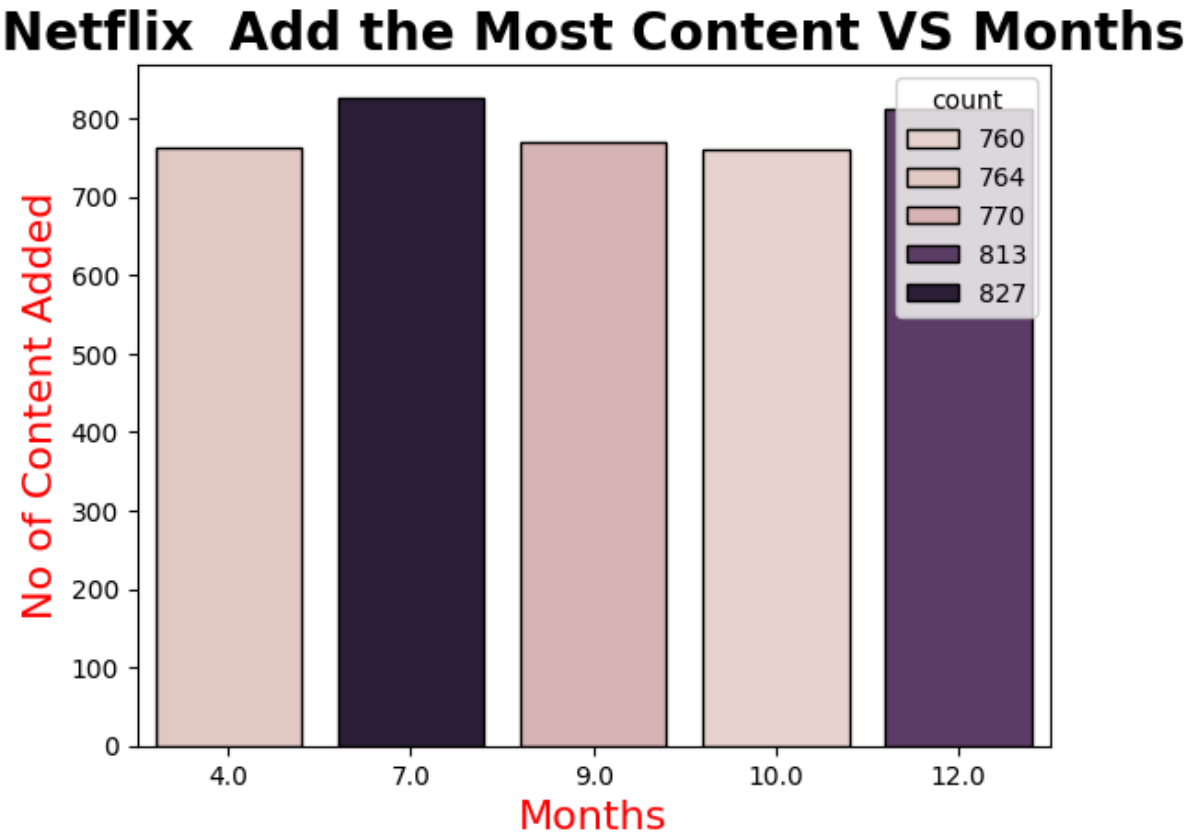
In [39]:
```
mov = net[net["type"]=="Movie"].copy()
mov["duration"]=mov["duration"].str.replace("min","").str.strip()
mov["duration"]=pd.to_numeric(mov["duration"],errors="coerce")
year=mov.groupby("release_year")["duration"].mean().sort_index(ascending=False)
year=year.head(10)
import matplotlib.pyplot as plt
import seaborn as sb
import matplotlib.pyplot as plt
sb.lineplot(x=year.index,y=year.values,marker="o",color="gray",linestyle="--")
plt.xlabel("Release Year")
plt.ylabel("Average Duration (minutes)")
plt.title("Trend of Movie Durations Over the Years")
plt.show()
```



Insight : Over the years, Netflix movies have been trending slightly shorter, edging closer to the 90-minute mark. This reflects audiences' shorter attention spans compared to the longer, 2-hour-plus epics of the early 2000s

18.In which months does Netflix add the most content?

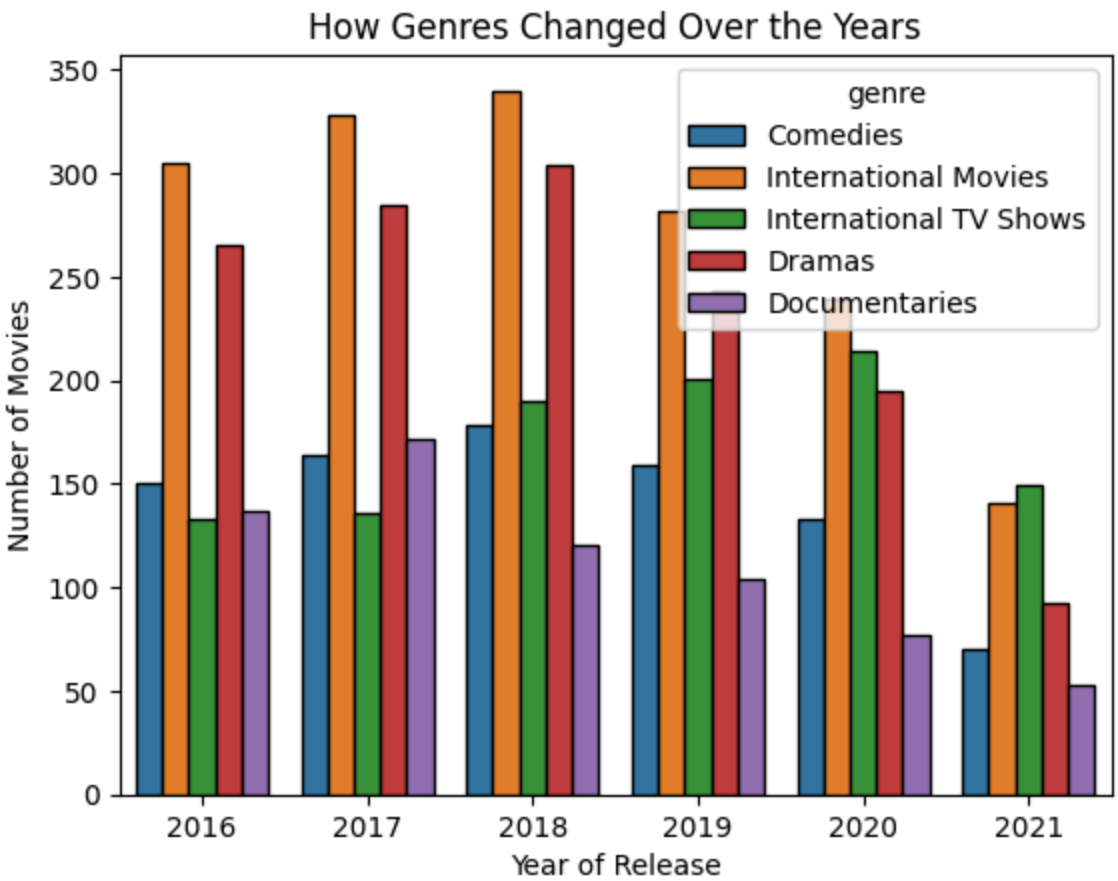
In [40]:
```
net["months"]=net["date_added"].dt.month
most_content=net["months"].value_counts().sort_values(ascending=False).head()
import matplotlib.pyplot as plt
import seaborn as sb
most_content
orders=["April","July","September","October","December"]
sb.barplot(x=most_content.index,y=most_content.values,hue=most_content,edgecolor="b
plt.xlabel("Months",color="red",fontsize=16)
plt.ylabel("No of Content Added",color="red",fontsize=16)
plt.title("Netflix  Add the Most Content VS Months",fontweight="bold",fontsize=20)
plt.show()
```



Insight : Netflix tends to add the most content in July, December, and January, which aligns with summer and winter holiday periods when global viewership is at its highest.

19. How does the genre distribution vary across different years?

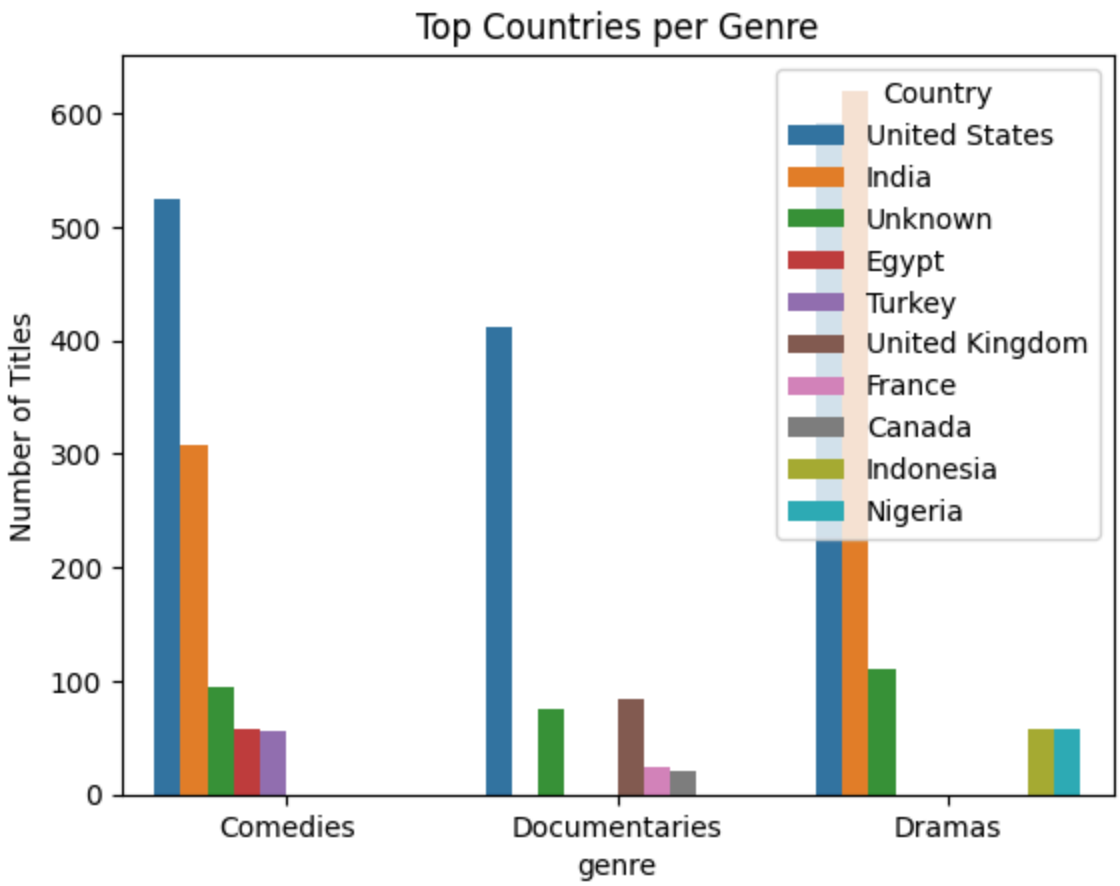
In [41]:
```
import matplotlib.pyplot as plt
import seaborn as sb
net['genre'] = net['listed_in'].str.split(', ')
genre_df = net.explode('genre')
top = genre_df['genre'].value_counts().head(5).index
recent_data = genre_df[(genre_df['genre'].isin(top)) & (genre_df['release_year'] >
sb.countplot(data=recent_data, x='release_year', hue='genre',edgecolor="black")
plt.title('How Genres Changed Over the Years')
plt.xlabel('Year of Release')
plt.ylabel('Number of Movies')
plt.show()
```



Insight : Over time, Netflix has evolved from mainly hosting comedies and dramas to placing a stronger focus on original documentaries and reality TV

20. Which countries produce the most content in each genre?

In [42]:
```
genre_country = net.assign(genre=net["listed_in"].str.split(",")).explode("genre")
genre_country["genre"] = genre_country["genre"].str.strip()
country_genre = genre_country.groupby(["country","genre"]).size().reset_index(name=
top_country_genre = country_genre.sort_values(["genre","count"], ascending=[True,Fa
top_country_genre = top_country_genre.groupby("genre").head(5)
top_country_genre.head()
import matplotlib.pyplot as plt
import seaborn as sb
target_genres = ['Dramas', 'Comedies', 'Documentaries']
sub_set = top_country_genre[top_country_genre['genre'].isin(target_genres)]
sb.barplot(data=sub_set, x='genre', y='count', hue='country')
plt.title("Top Countries per Genre")
plt.ylabel("Number of Titles")
plt.legend(title="Country", loc='upper right')
plt.show()
```



Insight : India produces a large number of dramas and comedies, while the United States leads in documentaries and horror. This highlights an opportunity for Netflix to focus on geo-targeted licensing by sourcing content that each region does bes

In [43]:
```
net.to_excel("net.xlsx")
```